

CSA 0378-DATA STRUCTURES

12. QUEUE USING ARRAY

```
main.c
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int queue[MAX];
6 int front = -1, rear = -1;
7
8 void enqueue(int value)
9 {
10     if(rear == MAX - 1)
11     {
12         printf("Queue Overflow\n");
13         return;
14     }
15
16     if(front == -1)
17         front = 0;
18
19     queue[++rear] = value;
20     printf("%d inserted into queue\n", value);
21 }
22
23 void dequeue()
24 {
25     if(front == -1 || front > rear)
26     {
27         printf("Queue Underflow\n");
28         return;
29     }
30
31     printf("Deleted Element: %d\n", queue[front++]);
32
33     if(front > rear)
34     {
35         front = -1;
36         rear = -1;
37     }
38 }
```

Output

```
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 1
Enter Value: 6
6 inserted into queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 1
Enter Value: 1
1 inserted into queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 2
Deleted Element: 6

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 1
Enter Value: 3
3 inserted into queue

1. Enqueue
2. Dequeue
3. Display
```

```
main.c
37 }
38 }
39
40 void display()
41 {
42     int i;
43
44     if(front == -1)
45     {
46         printf("Queue is Empty\n");
47         return;
48     }
49
50     printf("Queue Elements:\n");
51
52     for(i = front; i <= rear; i++)
53         printf("%d ", queue[i]);
54
55     printf("\n");
56 }
57
58 int main()
59 {
60     int choice, value;
61
62     while(1)
63     {
64         printf("\n1. Enqueue\n");
65         printf("2. Dequeue\n");
66         printf("3. Display\n");
67         printf("4. Exit\n");
68         printf("Enter Choice: ");
69         scanf("%d", &choice);
70
71         if(choice == 1)
72         {
73             printf("Enter Value: ");
74             scanf("%d", &value);
75             enqueue(value);
76
77         }
78         else if(choice == 2)
79         {
80             dequeue();
81
82         }
83         else if(choice == 3)
84         {
85             display();
86
87         }
88         else if(choice == 4)
89         {
90             break;
91         }
92         else
93         {
94             printf("Invalid Choice\n");
95         }
96     }
97     return 0;
98 }
```

Output

```
2. Dequeue
3. Display
4. Exit
Enter Choice: 1
Enter Value: 1
1 inserted into queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 2
Deleted Element: 6

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 1
Enter Value: 3
3 inserted into queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 3
Queue Elements:
1 3

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: |
```

```
main.c
61
62 while(1)
63 {
64     printf("\n1. Enqueue\n");
65     printf("2. Dequeue\n");
66     printf("3. Display\n");
67     printf("4. Exit\n");
68     printf("Enter Choice: ");
69     scanf("%d", &choice);
70
71     if(choice == 1)
72     {
73         printf("Enter Value: ");
74         scanf("%d", &value);
75         enqueue(value);
76
77     }
78     else if(choice == 2)
79     {
80         dequeue();
81
82     }
83     else if(choice == 3)
84     {
85         display();
86
87     }
88     else if(choice == 4)
89     {
90         break;
91     }
92     else
93     {
94         printf("Invalid Choice\n");
95     }
96 }
```

Output

```
2. Dequeue
3. Display
4. Exit
Enter Choice: 1
Enter Value: 1
1 inserted into queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 2
Deleted Element: 6

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 1
Enter Value: 3
3 inserted into queue

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: 3
Queue Elements:
1 3

1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter Choice: |
```